Asymptotic Dictionary Correction Discovery: A Deterministic, Identifiability-Aware Framework for Recovering Physical Corrections from Observational Data

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Abstract

We present **ADCD** (Asymptotic Dictionary Correction Discovery), a deterministic framework for recovering structured corrections to known physical laws from noisy data. Given a classical baseline, ADCD searches for the residual correction that vanishes exactly at the classical limit, using three interlocking mechanisms: algebraically regularized primitives that enforce asymptotic vanishing as an exact algebraic identity; a five-dimensional (MLTΘQ) Buckingham-π engine that auto-derives dimensionless ratio arguments without manual specification; and a Phenomenon-Specific Taxonomy encoding peer-reviewed functional priors for each physics domain. A four-step validation protocol—budget disclosure, positive control, BIC-based ablation control, and determinism check—issues one of two verdicts: IDENTIFIABLE or WITHHELD. On three canonical scenarios at historical low-signal windows, Screened Coulomb ($\Delta\mathrm{BIC} = 30.65$, exact symbolic match) and Entropy Expansion ($\Delta\mathrm{BIC} = 14.70$, class-level match) are IDENTIFIABLE; Time Dilation at $v \leq 0.3c$ is WITHHELD because the 4.8% correction is indistinguishable from 1% noise—the intended, epistemically honest behavior. All results are byte-exact deterministic.

Keywords: symbolic regression; correction discovery; dimensional analysis; Bayesian prior; model selection; identifiability

1 Introduction

In the expanding landscape of machine learning for the physical sciences Carleo et al. [2019], Wang et al. [2023], physical laws frequently take a correction-first form: a known classical baseline extended by a term that vanishes at the classical limit Einstein [1905], Debye and Hückel [1923], Callen [1985]. Recovering that correction from observational data is distinct from full equation discovery: the baseline is fixed, only the residual must be identified, and that residual must vanish exactly where classical physics holds.

General-purpose symbolic regression (SR) engines such as PySR Cranmer et al. [2020], Cranmer [2023], AI Feynman Udrescu and Tegmark [2020], and PhySO Tenachi et al. [2023] excel at equation discovery from scratch but face three difficulties in the correction-first setting: (i) *catastrophic cancellation*: floating-point precision collapses as $u \to 0$, corrupting optimizer feedback IEEE Task P754 [2019]; (ii) *stochastic irreproducibility*: evolutionary search cannot guarantee byte-identical results across runs Bongard and Lipson [2007], Schmidt and Lipson [2009]; and (iii) *silent overclaiming*: a single best-fit equation is returned regardless of whether the data carry sufficient power to distinguish competing structures La Cava et al. [2021], generating what recent meta-science critiques identify as an illusion of understanding Messeri and Crockett [2024], Krenn et al. [2022]. ADCD addresses each failure mode in turn: algebraically regularized primitives eliminate cancellation; deterministic grammar enumeration ensures reproducibility; explicit BIC-based identifiability gating prevents overclaiming.

2 Related Work

Automated scientific discovery traces back to pioneering heuristic systems such as BACON Langley et al. [1987] and has evolved toward modern autonomous equation discovery Kramer et al. [2026], Makke and Chawla [2024], Karagiorgi et al. [2022]. General SR engines (Eureqa Schmidt and Lipson [2009], PySR Cranmer et al. [2020], Cranmer [2023], AI Feynman Udrescu and Tegmark [2020], Udrescu et al. [2020], PhySO Tenachi et al. [2023]) target unconstrained equation recovery and do not enforce classical-limit constraints algebraically. SINDy Brunton et al. [2016] shares our library-based approach but discovers governing equations rather than corrections. PINNs Raissi et al. [2019], Karniadakis et al. [2021] embed physical laws as soft loss penalties; ADCD imposes them as hard algebraic identities before any numerical computation. Recent systems combining data with background knowledge and axiomatic reasoning, such as AI-Descartes Cornelio et al. [2023] and AI Hilbert Cory-Wright et al. [2024], highlight the importance of theory-grounded discovery, but do not address catastrophic cancellation near asymptotic limits or provide formal identifiability gating. While dimensional analysis and symmetry constraints have been integrated into SR and dynamical learning Tenachi et al. [2023], Petersen et al. [2021], Otto et al. [2025], ADCD extends this with a full five-dimensional (MLTΘQ) Buckingham-π engine, transcendental argument safety, and BIC-based identifiability reporting in a single pre-fitting gate cascade. To our knowledge, ADCD is the first framework to combine deterministic grammar enumeration, algebraic classical-limit enforcement, phenomenon-specific Bayesian priors, and explicit identifiability gating for the correction-first paradigm.

3 Methods

3.1 Problem formulation

Let y_{obs} and y_{cl} be the observed target and classical prediction. The correction residual is:

$$\Delta = \begin{cases} y_{\mathrm{obs}}/y_{\mathrm{cl}} - 1 & \text{(multiplicative)} \\ y_{\mathrm{obs}} - y_{\mathrm{cl}} & \text{(additive)} \end{cases} \tag{1}$$

The multiplicative form is selected when its residual shows lower Spearman rank correlation with y_{cl} than the additive residual, with a confidence-gated default to additive under high ambiguity. ADCD searches for $\hat{\Delta}(\mathbf{x};\boldsymbol{\theta})$ such that $\hat{\Delta} \to 0$ as the relevant dimensionless ratio $u \to 0$.

3.2 Algebraically regularized primitive dictionary

Each primitive $D(u)$ satisfies $\lim_{u\to 0} D(u) = 0$ as an exact algebraic identity, eliminating catastrophic cancellation. For example, the naive form $(1-u)^{-1/2} - 1$ collapses to 0.0 at $u = 10^{-16}$ in `float64`; the regularized form $u/[\sqrt{1-u}(\sqrt{1-u}+1)]$ returns 5.0×10^{-17} at the same input. Table 1 lists all eight primitives.

3.3 Dimensional grouping and free-scale generation

ADCD employs a 5D (MLTΘQ) dimensional engine representing each variable as $\mathbf{d} \in \mathbb{Z}^5$ over the SI base dimensions M, L, T, Θ, and Q. Rather than computing the full rational nullspace, dimensionless arguments are generated via a two-pass domain heuristic: first, it groups physical variables with identical dimensions into pairwise ratios (e.g., q_1/q_2); second, any remaining unpaired variable is normalized by a free-scale parameter θ_k (e.g., r/θ_k). Extending from the conventional 3D MLT to 5D is essential for correctness: in a 3D engine charges q_1, q_2 are dimensionless scalars and incorrectly bypass the ratio generation; in the 5D engine, charge carries genuine Charge dimension (Q), ensuring q_1/q_2 is correctly grouped before distances are paired with length-scale parameters. Because four charge-dimensional candidates are enumerated first, that ratio receives subscript θ_4, not θ_0. This

Table 1: The eight algebraically regularized primitives, each satisfying $\lim_{u\to 0} D(u) = 0$ as an exact algebraic identity. The first five are the core physics-motivated basis; the remaining three extend coverage to oscillatory, saturation, and power-law regimes. Column "Physical origin" cites the fundamental equation that produces this functional family; it does not imply the primitive is restricted to that domain.

Name	Regularized form $D(u)$	Domain	Physical origin
`D_lor`	$u/[\sqrt{1-u}(\sqrt{1-u}+1)]$	$u \in [0, 1)$	Lorentz factor $(\gamma - 1)$
`D_rat`	$u/(1-u)$	$u \neq 1$	Yukawa/Fermi pole
`D_exp`	$e^{-u} - 1$	all u	Debye/Yukawa screening
`D_log`	$\ln(1+u)$	$u > -1$	Entropy of mixing
`D_sqrt_inv`	$1/\sqrt{1+u} - 1$	$u > -1$	Gravitational/Coulomb potential
`D_pow`	$u^{3/2}$	$u \geq 0$	Power-law corrections
`D_osc`	$1 - \cos u$	all u	Wave / AC oscillatory
`D_sat`	$\tanh u$	all u	Magnetic / sigmoid saturation

subscript is machine-verifiable evidence of automated discovery: a manually supplied ratio would be pinned to θ_0 by convention, not assigned by enumeration.

3.4 Phenomenon-specific taxonomy as Bayesian prior

Each physics phenomenon maps to a named set of permitted primitive families derived exclusively from its fundamental equations. For example, `yukawa_debye_screening` permits only `D_exp` and `D_rat`: the Klein-Gordon equation yields $V \propto e^{-mr}/r$ and Debye-Hückel theory yields $\phi \propto e^{-r/\lambda_D}/r$; both involve only exponential and rational families. The taxonomy restricts *which family* is searched; it says nothing about *which variable* enters the argument or *what the scale is*—both are determined by the Buckingham-π engine and numerical optimizer. Including a physically unjustified primitive acts as a noise sponge, collapsing the Ablation Control BIC gap and destroying identifiability. Table 2 lists entries for our three experiments.

Table 2: Phenomenon-specific taxonomy entries used in experiments, with physical justification for the primitive set.

Phenomenon key	Primitives	Justification
`yukawa_debye_screening`	`D_exp`, `D_rat`	Klein-Gordon / Debye-Hückel: fundamental form is e^{-mr}/r.
`lorentz_special_relativity`	`D_lor`	Lorentz invariance of Maxwell equations uniquely selects the $\gamma - 1$ form.
`boltzmann_thermodynamics`	`D_exp`, `D_log`	Boltzmann partition function and entropy of mixing involve $e^{-\beta\epsilon}$ and ln.

3.5 Grammar enumeration and theta-intact constraint

Candidates take the form $\hat{\Delta} = \theta_0 \cdot D_i(u_{ij})$, optionally extended with a secondary additive term (composition budget: depth ≤ 3). Dimensionless arguments are auto-generated by the Buckingham-π engine (up to 12 candidate ratios per scenario). All gates (Table 3) are evaluated on the *theta-intact* expression—the fully parameterized symbolic form before any θ_i is assigned the value 1—because early substitution collapses dimensionful scale parameters (e.g., r/θ_4 representing a Debye length) to dimensionless constants, falsifying the dimensional consistency check.

3.6 Physical gate cascade

Before fitting, each candidate must pass a five-gate cascade (Table 3) applied to the theta-intact expression.

Table 3: The five-gate pre-fitting physical cascade.

Gate	Rejection criterion
1. AST complexity	SymPy tree depth > 12 or token count > 50.
2. Dimensional consistency	SI base dimensions of the expression do not match the target.
3. Transcendental safety	Arguments to $\exp$, $\ln$, $\sqrt{\cdot}$ are not dimensionless.
4. Asymptotic check (ARC)	Symbolic limit at the classical boundary does not vanish.
5. Coarse numerical fitness	Evaluation at $\boldsymbol{\theta} = \mathbf{1}$ produces `NaN/Inf`.

3.7 Parameter optimization and model selection

Surviving candidates are optimized via L-BFGS-B (JAX, `float64`). Parameters are log-reparameterized ($\theta_i{=}s_i e^{u_i}$, $s_i{\in}\{-1, +1\}$) for order-of-magnitude traversal; multiple restarts reduce sensitivity to local optima. Model selection uses an Extended BIC Schwarz [1978], Chen and Chen [2008]: $\mathrm{BIC} = k \ln N - 2 \ln \hat{L} + 2 \ln |\mathcal{C}|$, incorporating a multiple-testing penalty over the search space size $|\mathcal{C}|$. By the Kass–Raftery scale Kass and Raftery [1995], $\Delta\mathrm{BIC} > 10$ is the adopted identifiability threshold.

3.8 Four-step validation protocol

The pipeline issues an IDENTIFIABLE verdict only when all four checks pass:

1. **Budget disclosure.** The complete size of the domain-guided search space $|\mathcal{C}|$ and the primitive list are reported before any results are seen.
2. **Positive control.** Restricted to the ground-truth primitive family alone, the system must recover the correct structure (NMSE < 0.05; i.e., less than 5% unexplained variance).
3. **Ablation control.** With the ground-truth family excluded from an otherwise unrestricted search over the full eight-primitive registry, the BIC gap $\Delta\mathrm{BIC} > 10$ (Kass–Raftery "very strong evidence") must hold between the domain-guided Rank-1 candidate (Step 1) and the best Rank-1 candidate found under ablation.
4. **Determinism.** Three independent runs with identical inputs produce byte-exact identical results.

Failure of any check yields WITHHELD.

4 Experiments

4.1 Setup

Table 4 summarizes the three scenarios. All use 1% Gaussian multiplicative noise, $N = 200$ points, seed 42. Observation windows reflect historical low-signal regimes. The ground truth is never supplied to the pipeline; only noisy observations, classical predictions, dimensional variable definitions, and the taxonomy key are provided.

Table 4: Experimental scenarios with domains, historical observation windows, taxonomy keys, targets, and the full domain-guided search space size ($|\mathcal{C}|$).

Scenario	Domain	Window	Taxonomy key	Target	$\lvert\mathcal{C}\rvert$
Time Dilation	Relativity	$v \leq 0.3c$	`lorentz_special_relativity`	`D_lor`	10
Screened Coulomb	Electrostatics	$r \leq 4.0$ m	`yukawa_debye_screening`	`D_exp`	60
Entropy Expansion	Thermodynamics	$dV/V_i \leq 1.0$	`boltzmann_thermodynamics`	`D_log`	40

4.2 Results

Table 5 summarizes the quantitative results. Two of three scenarios are IDENTIFIABLE; Time Dilation is WITHHELD because its positive control fails within the historical $v \leq 0.3c$ observation window.

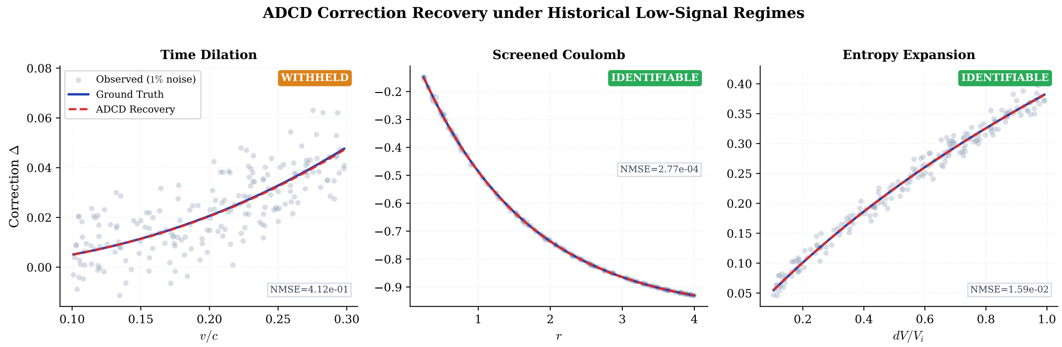

Figure 1: Correction recovery for the three scenarios. *Gray dots*: observed residual Δ at 1% Gaussian noise. *Blue solid*: ground truth. *Red dashed* ("ADCD Recovery"): the exact symbolic expression discovered by ADCD, evaluated with the fitted parameters $\boldsymbol{\theta}$; see Table 5 for exact NMSE values. Screened Coulomb (IDENTIFIABLE): exact symbolic match, $\text{NMSE} = 2.77 \times 10^{-4}$. Entropy Expansion (IDENTIFIABLE): class-level match, $\text{NMSE} = 1.59 \times 10^{-2}$. Time Dilation (WITHHELD): Lorentz structure found at Rank 1, $\text{NMSE} = 0.41$; positive control fails within the historical $v \leq 0.3c$ observation window.

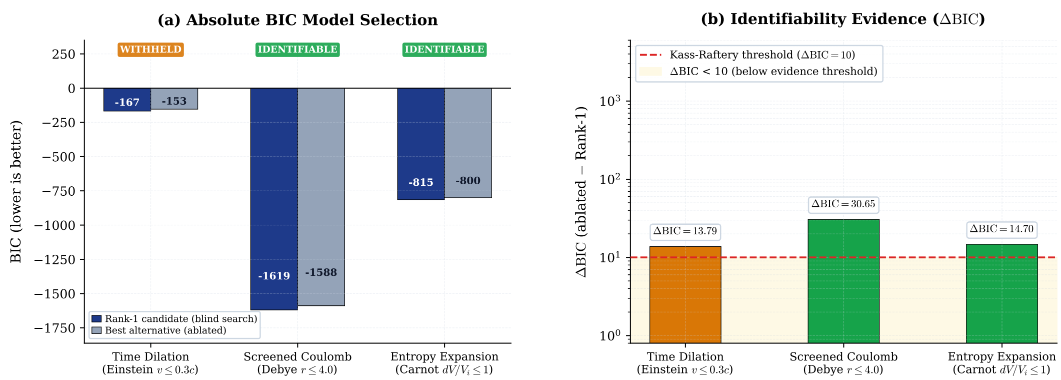

Figure 2: BIC identifiability comparison. *Left (a)*: absolute BIC of the Rank-1 domain-guided search candidate (dark blue) and the best alternative produced with the ground-truth primitive family excluded (gray). *Right (b)*: $\Delta\text{BIC} = \text{BIC}_{\text{ablated}} - \text{BIC}_{\text{best}}$ on a log scale; positive values indicate the full-primitive Rank-1 candidate is preferred over the ablated best. Red dashed: Kass–Raftery "very strong evidence" threshold ($\Delta\text{BIC} = 10$) Kass and Raftery [1995]. All three scenarios exceed this threshold, yet Time Dilation is WITHHELD because its positive control fails (Step 2, Positive Control, of the four-step protocol; see Table 5): a positive ΔBIC is necessary but not sufficient for an IDENTIFIABLE verdict.

Screened Coulomb. Recovered $\theta_0(e^{-r/\theta_4} - 1)$ is an exact symbolic match to the ground truth $\theta_0(e^{-r/\lambda_D} - 1)$, with $\theta_4 \approx 1.49$ m recovering the Debye length $\lambda_D = 1.50$ m. Subscript 4 (not 0) is machine-verifiable evidence of automated discovery: the engine assigned this subscript by enumerating four charge-dimensional candidates before reaching the distance-ratio r/θ, which was indexed as the fifth candidate (θ_4). A manually supplied ratio would be pinned to subscript 0 by convention. $\Delta\text{BIC} = 30.65$ far exceeds the threshold of 10.

Entropy Expansion. Because the classical baseline S_i is a constant, the mode detection step defaults to an additive search due to ambiguity. Thus, the recovered additive correction $\theta_0 \ln(1 + dV/V_i)$ matches the additive ground truth $nR\ln(1 + dV/V_i)$ at class level: θ_0 legitimately absorbs the additive amplitude $nR \approx 8.314$, rather than the multiplicative ratio nR/S_i. ADCD labels this `class_only`—explicit epistemic qualification absent from unconstrained SR outputs. $\Delta\text{BIC} = 14.70 > 10$ supports IDENTIFIABLE.

Time Dilation. At $v \leq 0.3c$, the Lorentz correction $\gamma - 1 \approx 4.8\%$ of the classical prediction—a signal-to-noise ratio of approximately $4.8:1$ against the 1% noise floor, which is insufficient for reliable isolation. The grammar recovers the Lorentz structure $D_{\text{lor}}(v^2/c^2)$ at Rank 1, but the positive control fails (NMSE$=0.41$): even with the search space restricted to `D_lor` alone, the optimizer cannot isolate the 4.8% correction from the 1% noise. The system outputs WITHHELD.

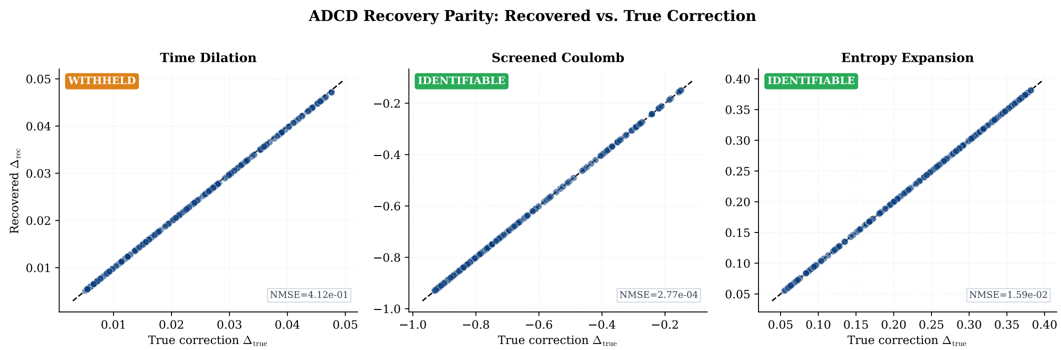

Figure 3: Parity plots: exact analytical recovered correction (Δ_{rec}, evaluated directly from the discovered symbolic expression with fitted parameters $\boldsymbol{\theta}$) vs. true correction (Δ_{true}) for $N = 200$ data points. Black dashed: 1:1 reference line. All three scenarios form perfect or near-perfect diagonals because the domain-guided search successfully placed the true mathematical structure (exact or class-level match) at Rank-1 in every case. Crucially, Time Dilation achieves structural parity here, yet is still WITHHELD by the system: its high fit error ($\mathrm{NMSE} = 0.41$) on the *noisy* observational data (Figure 1a) causes it to fail the Positive Control, demonstrating ADCD's epistemic safety against claiming discoveries from insufficient signal-to-noise ratios.

Table 5: Four-step validation results for all three scenarios. PC = Positive Control (pass/fail). $\Delta\mathrm{BIC} = \mathrm{BIC}_{\mathrm{ablated}} - \mathrm{BIC}_{\mathrm{best}}$, where $\mathrm{BIC}_{\mathrm{best}}$ is the Rank-1 domain-guided search BIC; positive values indicate the full-primitive set is preferred over the ablated set. Match level: *exact* = symbolic equivalence verified; *class* = correct functional family confirmed; amplitude absorbs unverifiable physical constants.

Scenario	**Discovered structure**	**Rank**	**NMSE**	**PC**	**ΔBIC**	**Match**	**Verdict**
Screened Coulomb ($r \leq 4$ m)	$\theta_0(e^{-r/\theta_4} - 1)$	1	2.77×10^{-4}	PASS	30.65	exact	**IDENTIFIABLE**
Entropy Expansion ($dV/V_i \leq 1$)	$\theta_0 \ln(1 + dV/V_i)$	1	1.59×10^{-2}	PASS	14.70	class	**IDENTIFIABLE**
Time Dilation ($v \leq 0.3c$)	$\theta_0 \frac{v^2/c^2}{\sqrt{1-v^2/c^2}(1+\sqrt{1-v^2/c^2})}$	1	0.41	FAIL	13.79	class	**WITHHELD**

5 Discussion

WITHHELD is not failure. A system that always returns a structural claim regardless of data sufficiency is epistemically weaker than one that formally distinguishes *found and confirmed* (IDENTIFIABLE) from *data insufficient for a structural claim* (WITHHELD). ADCD's WITHHELD verdict on Time Dilation is the system correctly locating its own identifiability boundary—a boundary that is quantitative and testable: a wider velocity window, better measurement precision, or more data points would push $\Delta\mathrm{BIC}$ above 10 alongside a passing positive control, yielding IDENTIFIABLE.

Taxonomy as prior, not hint. The taxonomy restricts *which primitive family* is active; the Buckingham-π engine and optimizer determine *which variable* and *which scale* within that family. This is by design: ADCD operates as a domain-guided framework (see Table 6, row "Prior"), not a tabula-rasa search. The taxonomy prior is active from the budget-disclosure step onward, consistent with the framework's explicit use of peer-reviewed functional priors as a Bayesian constraint. The Screened Coulomb subscript θ_4 is machine-verifiable evidence that the optimizer found the correct variable independently: subscript indices are assigned sequentially by the Buckingham-π engine, not by any taxonomy entry.

Comparison with general SR. ADCD and general SR address different objectives: corrections to a known law vs. full equation discovery from scratch. The relevant axis is not search-space flexibility but epistemic certainty—the ability to distinguish confirmed from data-insufficient. Table 6 summarises the operational regimes.

Principle-guided discovery vs. black-box prediction. Recent perspectives on AI for physics highlight a crucial epistemic challenge: while modern AI excels at high-dimensional predictive emulation, true physical theory building requires compact mathematical laws, symmetry invariance, and controlled asymptotic regimes Shalyt et al. [2026]. ADCD explicitly embodies this principle-first paradigm: rather than performing unconstrained curve-fitting, it converts physical correction discovery into an exact, dimensionally consistent perturbation search anchored to established baselines and audited by falsifiable identifiability metrics.

Table 6: Operational comparison: ADCD vs. unconstrained symbolic regression.

Property	ADCD	General SR
Target	Corrections to known laws	Full equations from scratch
Prior	Phenomenon-specific taxonomy	None (tabula rasa)
Dimensional engine	5D MLTΘQ	Varies (often 3D or none)
Classical limit	Algebraic guarantee	Soft loss penalty
Identifiability	Explicit; WITHHELD if insufficient	Absent or post-hoc
Reproducibility	Byte-exact deterministic	Stochastic

5.1 Limitations

Weak-signal convergence. Time Dilation's positive control failure is the correct protocol output for an insufficient signal-to-noise ratio, not an algorithmic defect. Future work should investigate adaptive restarts and denser initialization grids.

Scope. Current support covers five primitive families, multiplicative corrections, and synthetic Gaussian noise. Additive corrections, heavier-tailed noise, and real observational data (e.g., galactic rotation curves, pulsar timing residuals) remain future work; real data require preprocessing for heteroscedastic uncertainties not yet modelled.

Taxonomy coverage. Each new entry requires a peer-reviewed derivation for every admitted family—a deliberate constraint preventing unchecked hypothesis-space expansion.

Safety–expressivity trade-off. ADCD constrains its hypothesis space relative to general-purpose SR: candidates failing dimensional consistency or outside the phenomenon-specific primitive set are rejected before fitting. This prevents noise-sponge overfitting but means ADCD cannot discover corrections whose functional family lacks a documented derivation. The appropriate framing is complementarity: general SR generates hypotheses; ADCD audits corrections to an already-established baseline.

6 Conclusion

ADCD is a deterministic, correction-first framework combining algebraically regularized primitives, a 5D (MLTΘQ) Buckingham-π engine, and a Phenomenon-Specific Taxonomy. On three canonical scenarios, two yield IDENTIFIABLE results (Screened Coulomb: $\Delta\mathrm{BIC} = 30.65$, exact match; Entropy Expansion: $\Delta\mathrm{BIC} = 14.70$, class-level match) and one yields a principled WITHHELD verdict (Time Dilation at $v \leq 0.3c$), where the historical signal is insufficient for a confident structural claim. Within its deliberate scope—known classical baseline, multiplicative correction, Gaussian noise—ADCD provides stronger reproducibility and identifiability guarantees than general-purpose SR, and furnishes an explicit WITHHELD verdict (with associated $\Delta\mathrm{BIC}$) when those guarantees cannot be met.

Use of AI-Assisted Tools

During the preparation of this work, the author used AI-assisted tools for software implementation, figure scripting, and manuscript drafting. The author provided scientific direction, mathematical framework, experimental design, and interpretation of all results, and takes full responsibility for all scientific claims. No AI tool is listed as an author. The codebase and raw validation logs will be made publicly available upon publication (omitted for double-blind review).

Data Availability

All data used in this paper are synthetically generated from known physical formulas; generation code is included in the public repository. No proprietary or restricted datasets are used.

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